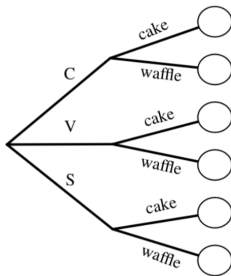
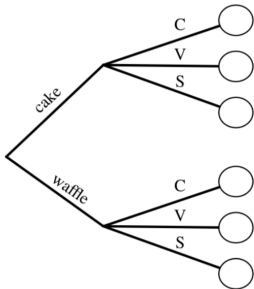


Sec 2.3 Counting

Definition (Multiplication Rule)

Consider a compound experiment consisting of two sub-experiments, Experiment A and Experiment B. Suppose that Experiment A has m possible outcomes, and for each of those outcomes Experiment B has n possible outcomes. Then the compound experiment has mn possible outcomes.

Eg 1.(from Blitzstein and Hwang stat book) Suppose you are buying an ice cream cone. You can choose whether to have a cake cone or a waffle cone, and whether to have chocolate, vanilla, or strawberry as your flavor. This decision process can be visualized with a tree diagram,



$$\frac{2}{\text{Cone}} \quad \frac{3}{\text{Flavour}} = 6$$

Generalized Multiplication Principle

Consider a compound experiment consisting of k sub-experiments, assume experiment k can occur in n_k possible ways, then the number of ways in which the sequence of k experiment may occur is

$$n_1 \cdot n_2 \cdots n_k$$

Eg 2. Number of ways to choose a license plate with 2 letters followed by 3 digits.

Assuming repetition is allowed.

$$\underline{26} \quad \underline{26} \quad \underline{10} \quad \underline{10} \quad \underline{10} = (26)^2 \cdot 10^3$$

Eg 3. A tech company has developed five AI models—Alpha and Beta (first-generation models) and Delta, Epsilon, and Gamma (their upgraded successors). These five models need to be distributed across seven different research labs. However, due to compatibility constraints, no lab can receive both a first-generation model and a successor model. Additionally, not every lab is required to receive an AI model.

In how many different ways can these AI models be distributed under these conditions?

There are two different scenarios,
either $S_1 =$ the two First generations are distributed
to the same lab
or
 $S_2 =$ the two First generations are distributed
to two different labs.

total possible ways = $|S_1| + |S_2|$

If the two first generation models are given to the same lab then it could be any of the 7 labs, then there are 6 labs left to distribute the 2nd generation models and there is no restriction for each model there are 6 possible choices.

$$|S_1| = \frac{7}{\substack{\text{Model } \alpha \text{ and} \\ \text{Model } \beta}} \frac{6}{\substack{\text{Model} \\ \text{delta}}} \frac{6}{\substack{\text{Model} \\ \text{Gamma}}} \frac{6}{\substack{\text{Model} \\ \text{Epsilon}}} = 7 \cdot 6^3$$

For scenario S_2 , the first generation model must be distributed to two different labs, so 7 possible scenarios, for model β 6 possible scenarios, and

the 2nd generations could be distributed to any remaining of 5 with no restriction.

$$|S_2| = \frac{7}{\text{Model } \alpha} \frac{6}{\text{Model } \beta} \frac{5}{\text{Model } \delta} \frac{5}{\text{Model } \gamma} \frac{5}{\text{Model } \epsilon} = 7 \cdot 6 \cdot 5^3$$

$$\text{Total} = |S_1| + |S_2| = 7 \cdot 6^3 + 7 \cdot 6 \cdot 5^3$$

"Eg3 a different version"

Let's say we have two adult dogs 1 female dog and 1 male dog and three puppies in a shelter.

There will be 7 people who visit the shelter today to adopt some dogs. Let's say we must not give an adult dog and a puppy to the same person. It's possible that some people get no dog. Also all the dogs must be adopted. How many different scenarios are possible.

The method is exactly the same as in Eg3.

Eg 4. How many ways are there to arrange 5 books on a bookshelf?

$$\frac{5}{1\text{st}} \frac{4}{2\text{nd}} \frac{3}{3\text{rd}} \frac{2}{4\text{th}} \frac{1}{5\text{th}} = 5!$$

Note: $n!$ is the number of ways to rearrange (reorder) n distinct items.

$$n! = 1 \cdot 2 \dots (n-1) \cdot n$$

$$n! = n \cdot (n-1) \dots 2 \cdot 1$$

$$0! = 1$$

Definition (Permutations)

Permutations are the possible ordered selections of k objects out of a total of n objects.

In fact, consider n objects and making k choices from them, one at a time without replacement (i.e., choosing a certain object precludes it from being chosen again). Then there are ${}^n P_k$ possible outcomes where

$${}^n P_k = \frac{n!}{(n-k)!} \quad (1)$$

We can also use the notation $P(n, k) = {}^n P_k$.

Eg 5. How many ways are possible to make a code of length 10 from natural numbers $1, 2, \dots, 100$, where no repetition is allowed.

It's a permutation of length $k=10$, from $n=100$.

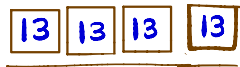
$$\text{answer } P(100, 10) = \frac{100!}{90!}$$

Eg 6. A standard 52-card deck consists of one of each combination of: 13 different ranks (Ace, 2, 3,...,10, Jack, Queen, King) and 4 different suits (clubs, diamonds, hearts, spades). In how many ways a 52-card deck be dealt to thirteen players, four to each, so that every player has one card of each suit?

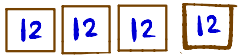
For solving this problem one approach will be to consider that for each suit [includes 13 cards] in how many ways we can give them to 13 players $\left[\frac{13}{p_1} \frac{12}{p_2} \frac{11}{p_3} \dots \frac{1}{p_{13}} = 13! \right]$.

This follows for each suit and we have 4 suits so in total $(13!)^4$.

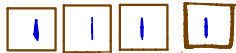
second way of thinking about this problem:



player 1



player 2



player 13

$$(13)^4 \cdot (12)^4 \cdot (11)^4 \cdot \dots \cdot (1)^4 = (13 \cdot 12 \cdot 11 \cdot \dots \cdot 1)^4 \\ = (13!)^4$$

Eg 7. How many different letter arrangements can be formed from the letters CACAO?

In total there 5 words so $5!$ arrangement but consider AACCO we can change the first 2 As but still its same word. so we have divide by over counting of number of arrangement of alike objects.

so there are $\frac{5!}{2!2!}$ arrangements.

Definition (Number of Sequences of objects that some of them are alike)

In general, the same reasoning as that used in Eg 6 shows that there are

$$\frac{n!}{n_1!n_2!\cdots n_r!} \quad (2)$$

different permutations of objects, of which n_1 are alike, n_2 are alike,... n_r are alike.

Definition (Combinations)

Combinations are the possible selections of k items from a group of n items regardless of the order of selection. The number of combinations of n objects taken k at a time is

$${}^nC_k = \binom{n}{k} = \frac{n!}{(n-k)!k!} = \frac{{}^nP_k}{k!} \quad (3)$$

Eg 8. A pizza store offers 6 different toppings from which one could select their pizza topping from. How many different three-topping pizzas can you get if each topping can be put on a pizza at most once? (The order in which the toppings are selected does not matter.)

since the order of the toppings doesn't matter,
we only need to choose 3 toppings. The answer
will be $\binom{6}{3}$ or $\frac{6!}{3!3!} = \frac{{}^6P_3}{3!}$

Eg 9: If 3 balls are "randomly drawn" from a bowl containing 6 white and 5 black balls, what is the probability that one of the balls is white and the other two black?

choosing 3 balls from $6w + 5b$,

$$|S| = \binom{11}{3}$$

$$\text{Event } A = 1w, 2b \Rightarrow |A| = \binom{6}{1} \binom{5}{2}$$

$$P(A) = \frac{\binom{6}{1} \binom{5}{2}}{\binom{11}{3}}$$

Eg 10: An urn contains n balls, one of which is special. If k of these balls are withdrawn, what is the probability that the special ball is chosen?

$$|S| = \binom{n}{k}$$

$$|E| = \binom{n-1}{k-1}$$

$$P(E) = \frac{|E|}{|S|} = \frac{\binom{n-1}{k-1}}{\binom{n}{k}} = \frac{k}{n}$$

Eg 11. (Problem 43 from Devore) In five-card poker, a straight consists of five cards with adjacent denominations (e.g., 9 of clubs, 10 of hearts, jack of hearts, queen of spades, and king of clubs). Assuming that aces can be high or low, if you are dealt a five-card hand,

- (a) What is the probability that it will be a straight with high card 10?
- (b) What is the probability that it will be a straight?
- (c) What is the probability that it will be a straight flush (all cards in the same suit)?

of total poker hands $\binom{52}{5}$

(a) meaning we have 10 9 8 7 6, since we already know the rank, we have to choose the suits [note that here order doesn't matter.]

$$\underline{4} \quad \underline{4} \quad \underline{4} \quad \underline{4} \quad \underline{4} = 4^5$$

$$\text{probability(high card 10)} = \frac{4^5}{\binom{52}{5}}$$

(b) straight means A 2 3 4 5, 2 3 4 5 6, ..., 10 J 9 K A,

we have 10 different case for the bottom number, or 10 different set of ranks that form a stright.

$$10 \quad \underline{4} \quad \underline{4} \quad \underline{4} \quad \underline{4} \quad \underline{4} = 10 \cdot 4^5$$

$$p(\text{straight}) = \frac{10 \cdot 4^5}{\binom{52}{5}}$$

(c) straight flush means it's a straight but all the cards are from same suit. Hence,

$$p(\text{straight flush}) = \frac{10 \cdot 4}{\binom{52}{5}}$$